

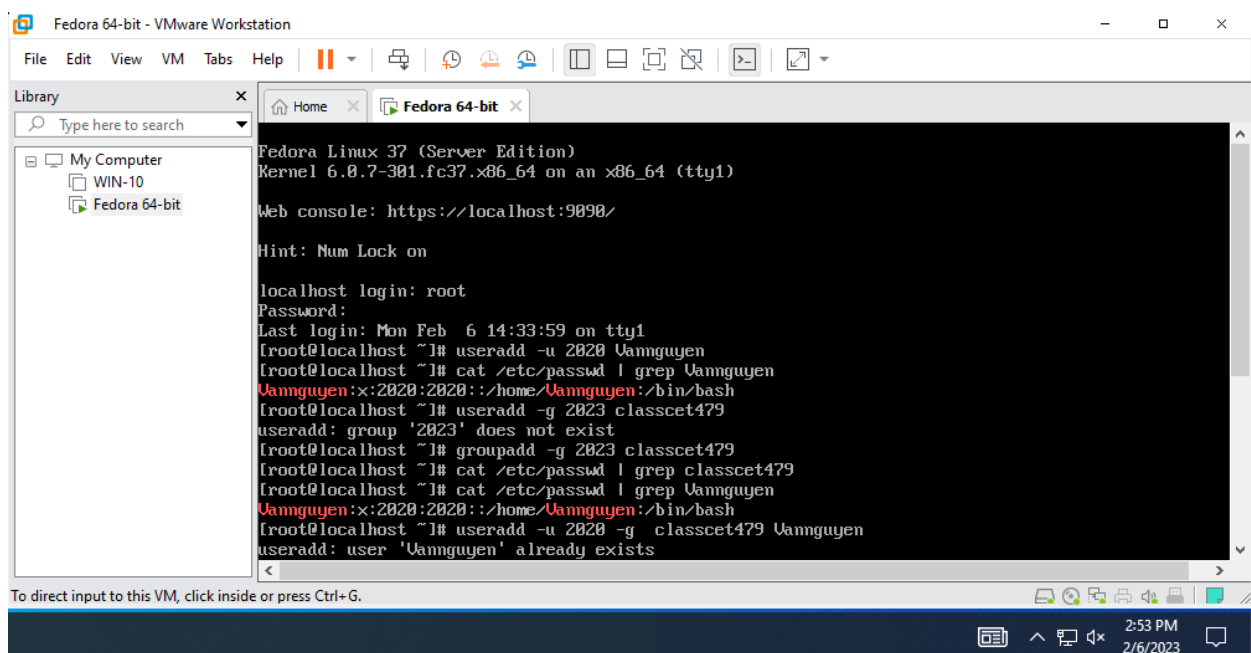
LAB_01 :

Commands in Fedora Linux

1. Adding **UID** (*Unique Identification Number*). and GID is done using which commands?

- Add UID [`useradd -u <UID> <user name>`]
- Add UID in a group : [`useradd -u <UID> -g <group name> <user name>`]
- Verify UID created [`cat /etc/passwd | grep <user name>`]
- # [`useradd <user name>`] : create a new user
- # [`passwd <user name>`] : create a password for new user
- # [`cat /etc/passwd`] : Check the password just created
- Add GID [`groupadd -g <group ID> <group name>`]
- Add GID in a group : [`useradd -u <UID> -g <group name> <user name>`]

Check if your proposed commands work by typing them in your Linux environment.



```
Fedora Linux 37 (Server Edition)
Kernel 6.0.7-301.fc37.x86_64 on an x86_64 (tty1)

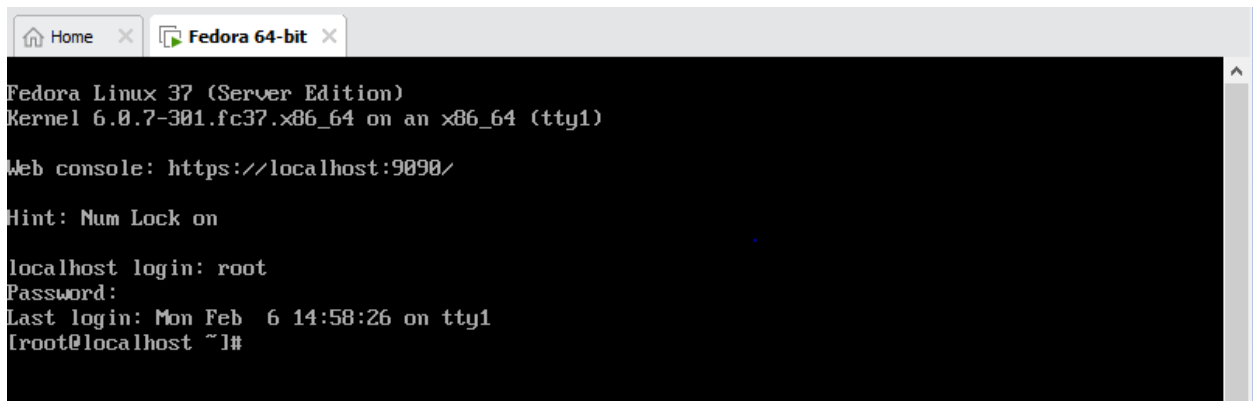
Web console: https://localhost:9090/

Hint: Num Lock on

localhost login: root
Password:
Last login: Mon Feb  6 14:33:59 on tty1
[root@localhost ~]# useradd -u 2020 Vannguyen
[root@localhost ~]# cat /etc/passwd | grep Vannguyen
Vannguyen:x:2020:2020::/home/Vannguyen:/bin/bash
[root@localhost ~]# useradd -g 2023 classcet479
useradd: group '2023' does not exist
[root@localhost ~]# groupadd -g 2023 classcet479
[root@localhost ~]# cat /etc/passwd | grep classcet479
[root@localhost ~]# cat /etc/passwd | grep Vannguyen
Vannguyen:x:2020:2020::/home/Vannguyen:/bin/bash
[root@localhost ~]# useradd -u 2020 -g classcet479 Vannguyen
useradd: user 'Vannguyen' already exists
```

2. By which command do you get the root access?

[root] or **[su -]**

A terminal window titled 'Fedora 64-bit' showing a login process. The text displayed is: 'Fedora Linux 37 (Server Edition)', 'Kernel 6.0.7-301.fc37.x86_64 on an x86_64 (tty1)', 'Web console: https://localhost:9090/', 'Hint: Num Lock on', 'localhost login: root', 'Password:', 'Last login: Mon Feb 6 14:58:26 on tty1', and '[root@localhost ~]#'.

```
Fedora Linux 37 (Server Edition)
Kernel 6.0.7-301.fc37.x86_64 on an x86_64 (tty1)

Web console: https://localhost:9090/

Hint: Num Lock on

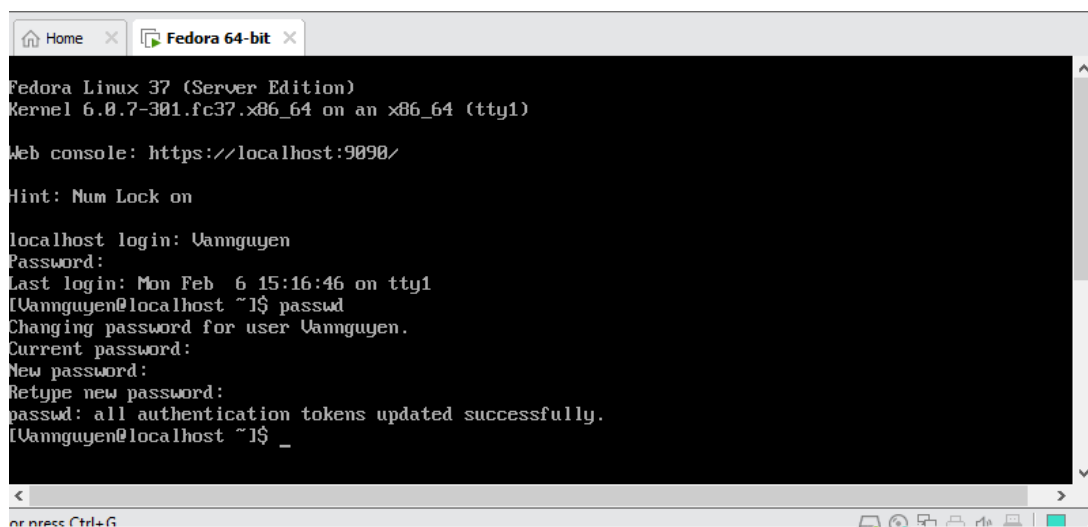
localhost login: root
Password:
Last login: Mon Feb 6 14:58:26 on tty1
[root@localhost ~]#
```

3. What are some known privileges of the root user?

The root account has root privileges. This means it can read and write any files on the system, perform operations as any user, change system configuration, install, and remove software, and upgrade the operating system and/or firmware.

4. Can the new users change their own password ? **Yes.**

\$[passwd] type the current password then type new password and retype again.

A terminal window titled 'Fedora 64-bit' showing a user logging in and changing their password. The text displayed is: 'Fedora Linux 37 (Server Edition)', 'Kernel 6.0.7-301.fc37.x86_64 on an x86_64 (tty1)', 'Web console: https://localhost:9090/', 'Hint: Num Lock on', 'localhost login: Vannguyen', 'Password:', 'Last login: Mon Feb 6 15:16:46 on tty1', '[Vannguyen@localhost ~]\$ passwd', 'Changing password for user Vannguyen.', 'Current password:', 'New password:', 'Retype new password:', 'passwd: all authentication tokens updated successfully.', and '[Vannguyen@localhost ~]\$ _'.

```
Fedora Linux 37 (Server Edition)
Kernel 6.0.7-301.fc37.x86_64 on an x86_64 (tty1)

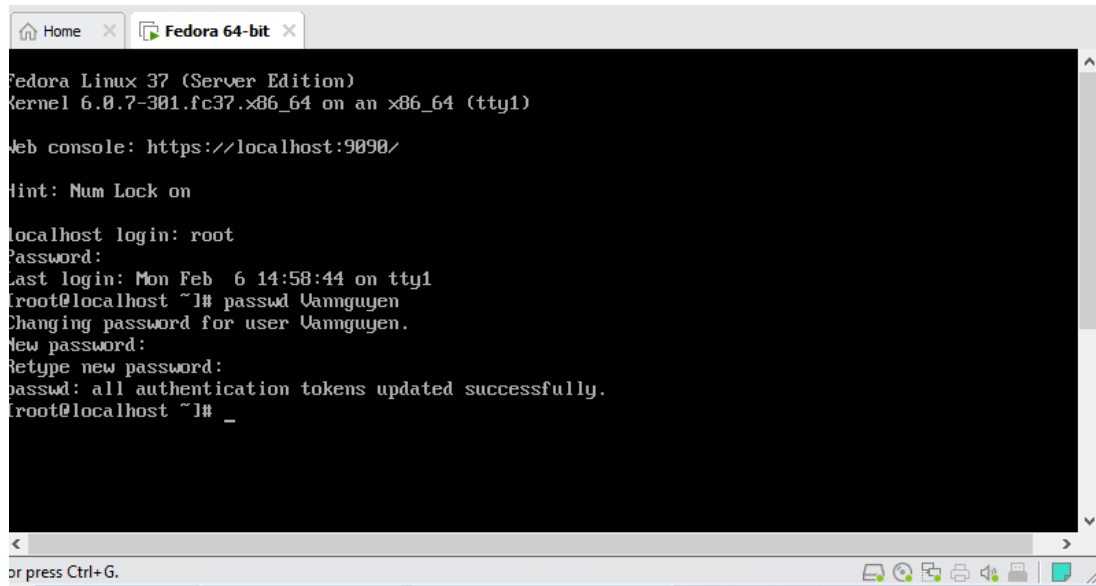
Web console: https://localhost:9090/

Hint: Num Lock on

localhost login: Vannguyen
Password:
Last login: Mon Feb 6 15:16:46 on tty1
[Vannguyen@localhost ~]$ passwd
Changing password for user Vannguyen.
Current password:
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[Vannguyen@localhost ~]$ _
```

5. Can the root change the password of the new users?

`$[ passwd <user name>]` type the new password then retype new password again.



A terminal window titled 'Fedora 64-bit' showing a root login session. The user 'root' logs in from 'localhost'. The command 'passwd Vannguyen' is executed to change the password for the user 'Vannguyen'. The terminal prompts for a new password and then for the password to be retyped. After successful verification, it displays 'passwd: all authentication tokens updated successfully.' and returns to the root prompt.

```
Fedora Linux 37 (Server Edition)
kernel 6.0.7-301.fc37.x86_64 on an x86_64 (tty1)

Web console: https://localhost:9090/

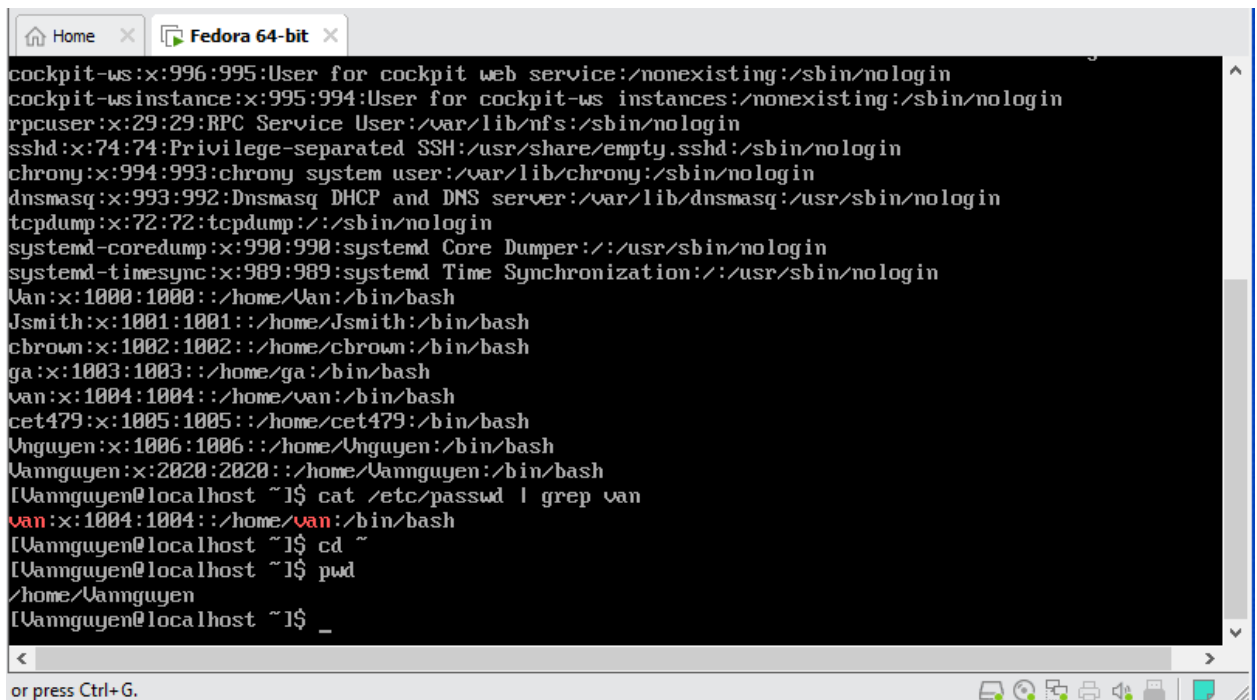
Hint: Num Lock on

localhost login: root
Password:
Last login: Mon Feb  6 14:58:44 on tty1
[root@localhost ~]# passwd Vannguyen
Changing password for user Vannguyen.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# _
```

6. If you want to check your current directory, which command will you use?

`[ pwd ]`

Check if your proposed command works.



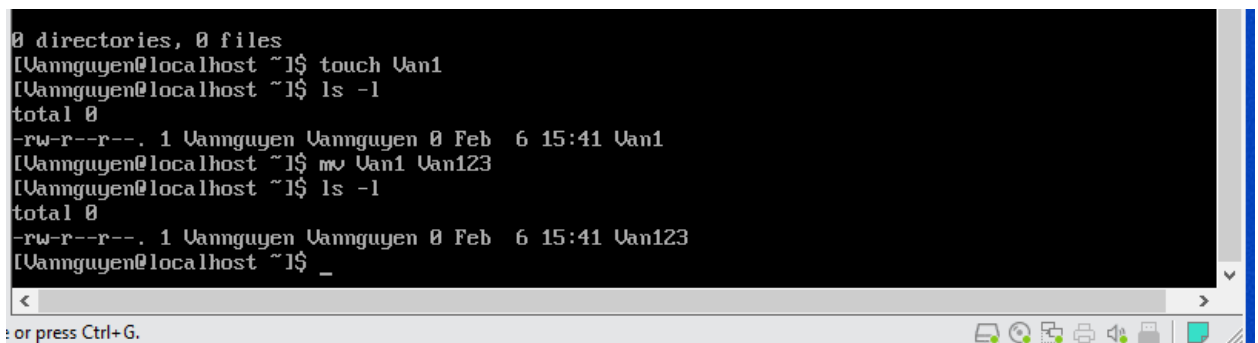
A terminal window titled 'Fedora 64-bit' showing a list of system users. The command 'cat /etc/passwd | grep van' is executed to filter users whose names contain 'van'. The output shows several users, including 'van' with UID 1004. The user then switches to the 'van' user using 'su - van' and runs the 'pwd' command to verify the current directory, which outputs '/home/Vannguyen'.

```
cockpit-ws:x:996:995:User for cockpit web service:/nonexisting:/sbin/nologin
cockpit-wsinstance:x:995:994:User for cockpit-ws instances:/nonexisting:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
chrony:x:994:993:chrony system user:/var/lib/chrony:/sbin/nologin
dnsmasq:x:993:992:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/usr/sbin/nologin
tcpdump:x:72:72:tcpdump:/:/sbin/nologin
systemd-coredump:x:990:990:systemd Core Dumper:/usr/sbin/nologin
systemd-timesync:x:989:989:systemd Time Synchronization:/usr/sbin/nologin
Van:x:1000:1000::/home/Van:/bin/bash
Jsmith:x:1001:1001::/home/Jsmith:/bin/bash
cbrown:x:1002:1002::/home/cbrown:/bin/bash
ga:x:1003:1003::/home/ga:/bin/bash
van:x:1004:1004::/home/van:/bin/bash
cet479:x:1005:1005::/home/cet479:/bin/bash
Unguyen:x:1006:1006::/home/Unguyen:/bin/bash
Vannguyen:x:2020:2020::/home/Vannguyen:/bin/bash
[Vannguyen@localhost ~]$ cat /etc/passwd | grep van
van:x:1004:1004::/home/van:/bin/bash
[Vannguyen@localhost ~]$ su - van
[Vannguyen@localhost ~]$ cd ~
[Vannguyen@localhost ~]$ pwd
/home/Vannguyen
[Vannguyen@localhost ~]$ _
```

7. If you want to rename an empty file in the same directory, which set of command would you use?

```
mv <old name file> <new name file>
```

Check if your proposed commands work.

A terminal window showing a series of commands and their outputs. The user is at a prompt [Vannguyen@localhost ~]. They first run 'touch Van1', then 'ls -l' which shows a file 'Van1' with permissions '-rw-r--r--' and size '1'. Next, they run 'mv Van1 Van123', and 'ls -l' again, which now shows 'Van123' with the same permissions and size. The terminal window has a standard Linux desktop environment at the bottom with icons for network, disk, and other system utilities.

```
0 directories, 0 files
[Vannguyen@localhost ~]$ touch Van1
[Vannguyen@localhost ~]$ ls -l
total 0
-rw-r--r--. 1 Vannguyen Vannguyen 0 Feb  6 15:41 Van1
[Vannguyen@localhost ~]$ mv Van1 Van123
[Vannguyen@localhost ~]$ ls -l
total 0
-rw-r--r--. 1 Vannguyen Vannguyen 0 Feb  6 15:41 Van123
[Vannguyen@localhost ~]$ _
```

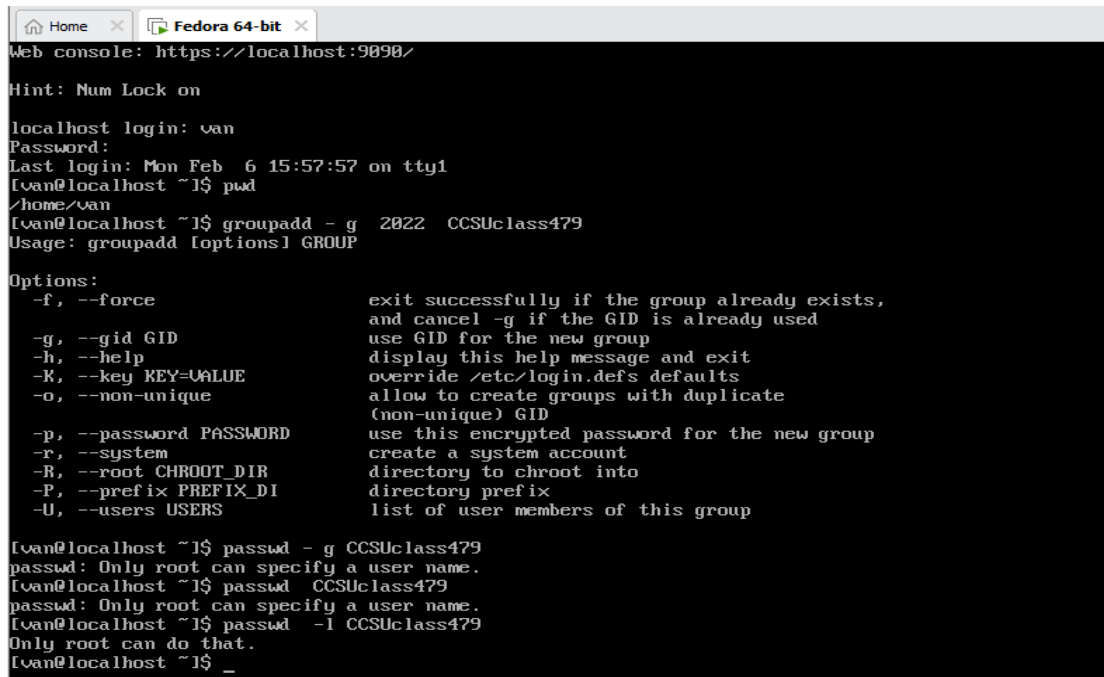
8. Create a new group in your home directory and change its password. Which set of commands did you use?

Check if your proposed commands work.

u

- **[groupadd -g]** <group-ID> <group-name>
- **[passwd -g]** <group-name>

The current group password is not prompted for. The -r option is used with the -g option to remove the current password from the named group. This allows group access to all members. The -R option is used with the -g option to restrict the named group for all users.



```

Web console: https://localhost:9090/

Hint: Num Lock on

localhost login: van
Password:
Last login: Mon Feb  6 15:57:57 on tty1
[van@localhost ~]$ pwd
/home/van
[van@localhost ~]$ groupadd -g 2022 CCSUclass479
Usage: groupadd [options] GROUP

Options:
  -f, --force                exit successfully if the group already exists,
                             and cancel -g if the GID is already used
  -g, --gid GID              use GID for the new group
  -h, --help                 display this help message and exit
  -k, --key KEY=VALUE        override /etc/login.defs defaults
  -o, --non-unique            allow to create groups with duplicate
                             (non-unique) GID
  -p, --password PASSWORD    use this encrypted password for the new group
  -r, --system               create a system account
  -R, --root CHROOT_DIR      directory to chroot into
  -P, --prefix PREFIX_DIR    directory prefix
  -U, --users USERS           list of user members of this group

[van@localhost ~]$ passwd -g CCSUclass479
passwd: Only root can specify a user name.
[van@localhost ~]$ passwd CCSUclass479
passwd: Only root can specify a user name.
[van@localhost ~]$ passwd -l CCSUclass479
Only root can do that.
[van@localhost ~]$ _

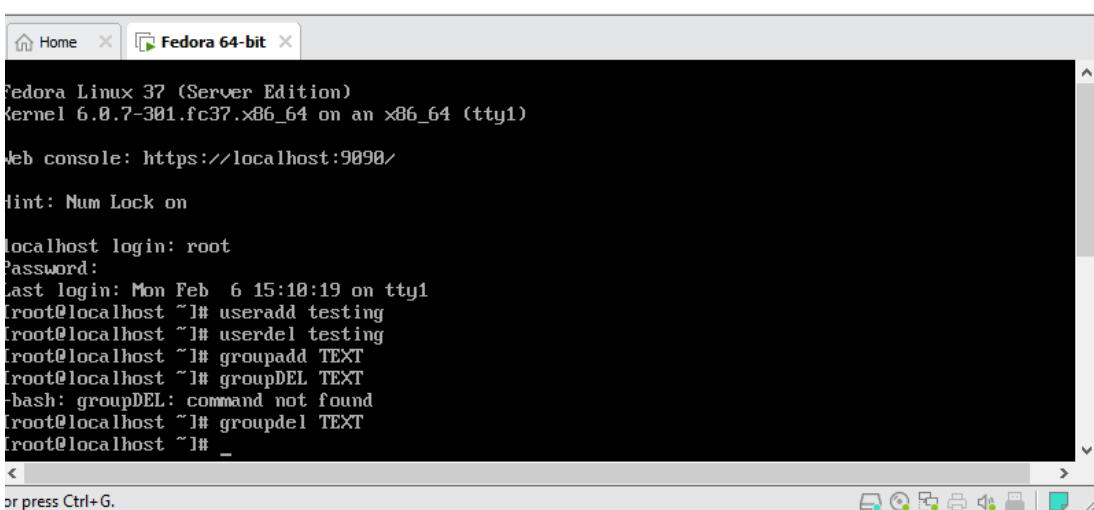
```

9. Now delete the user and group that you created. What set of commands did you use? Did your proposed commands work?

#[userdel <user name>] : Delete a user name user.

#[userdel -r <user name>] : delete the user's home directory and mail spool pass the **-r** option

#[groupdel <group name>] : Delete a group name.



```

Fedora Linux 37 (Server Edition)
Kernel 6.0.7-301.fc37.x86_64 on an x86_64 (tty1)

Web console: https://localhost:9090/

Hint: Num Lock on

localhost login: root
Password:
Last login: Mon Feb  6 15:10:19 on tty1
root@localhost ~# useradd testing
root@localhost ~# userdel testing
root@localhost ~# groupadd TEXT
root@localhost ~# groupDEL TEXT
-bash: groupDEL: command not found
root@localhost ~# groupdel TEXT
root@localhost ~# _

```

10. Which command would you use if you want to display the names of all files and directories in your current directory? Check if your proposed command works.

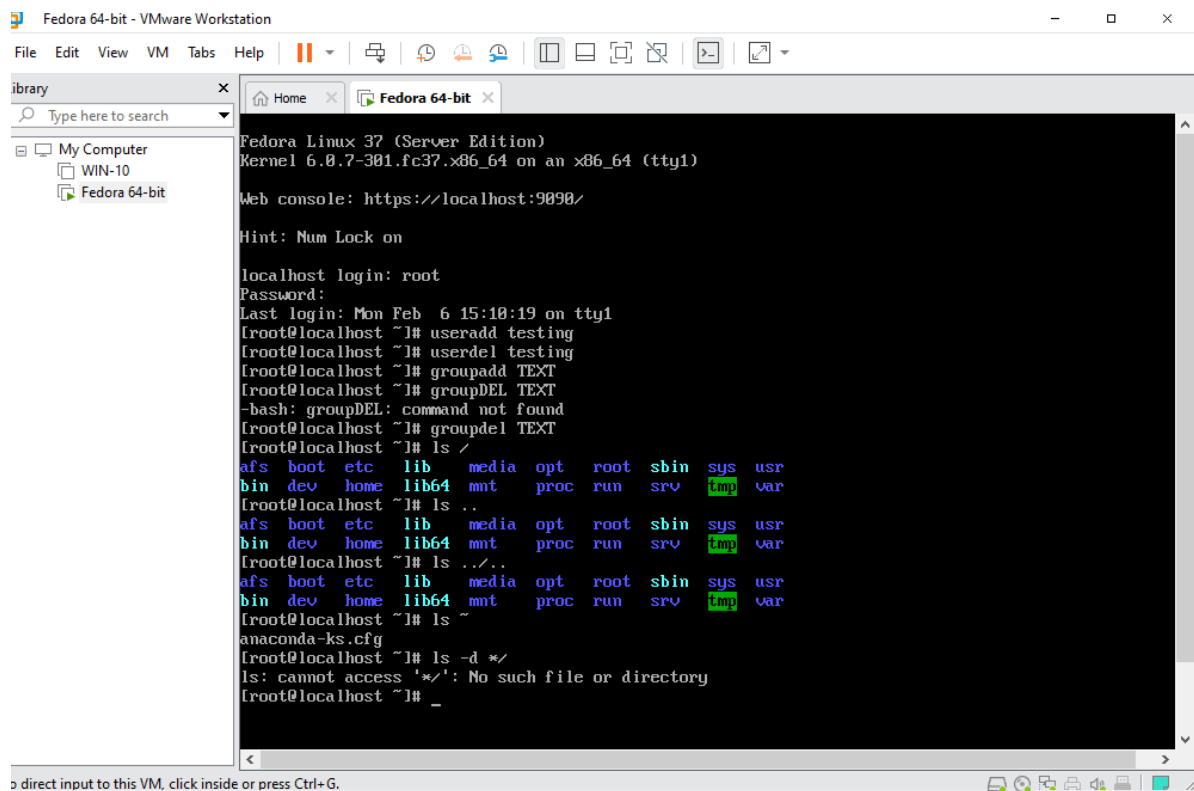
`#[ ls / ]` : to list the contents of the root directory.

`#[ ls .. ]` : to list the contents of the parent directory one.

`#[ ls ../../ ]` : to list for contents two levels above

`#[ ls ~ ]` : to list for contents in the users's home directory

`#[ ls -d */ ]` : to list to list only directories.



The screenshot shows a VMware Workstation window titled "Fedora 64-bit - VMware Workstation". Inside the window is a terminal window titled "Fedora Linux 37 (Server Edition)" with the following text:

```
Fedora Linux 37 (Server Edition)
Kernel 6.0.7-301.fc37.x86_64 on an x86_64 (tty1)

Web console: https://localhost:9090/

Hint: Num Lock on

localhost login: root
Password:
Last login: Mon Feb 6 15:10:19 on tty1
[root@localhost ~]# useradd testing
[root@localhost ~]# userdel testing
[root@localhost ~]# groupadd TEXT
[root@localhost ~]# groupDEL TEXT
-bash: groupDEL: command not found
[root@localhost ~]# groupdel TEXT
[root@localhost ~]# ls /
afs boot etc lib media opt root sbin sys usr
bin dev home lib64 mnt proc run srv tmp var
[root@localhost ~]# ls ..
afs boot etc lib media opt root sbin sys usr
bin dev home lib64 mnt proc run srv tmp var
[root@localhost ~]# ls ../../
afs boot etc lib media opt root sbin sys usr
bin dev home lib64 mnt proc run srv tmp var
[root@localhost ~]# ls ~
anaconda-ks.cfg
[root@localhost ~]# ls -d */
ls: cannot access '*/': No such file or directory
[root@localhost ~]# _
```

At the bottom of the terminal window, there is a status bar that reads: "direct input to this VM, click inside or press Ctrl+G."

11. Display the use of * wildcard in your Linux environment.

```
# [ls *.* ] : Show all files in current working directory.  
# [ls *.txt ] : Show all files that have the extension .txt  
# [ls ccsu *.*] : Show all files that have first four  
characters is ccsu.
```

12. Can you create three empty files test1.txt, test2.txt, and test3.txt using a single command? Show that it works.

```
# touch test1.txt  
# touch test2.txt  
# touch test3.txt
```

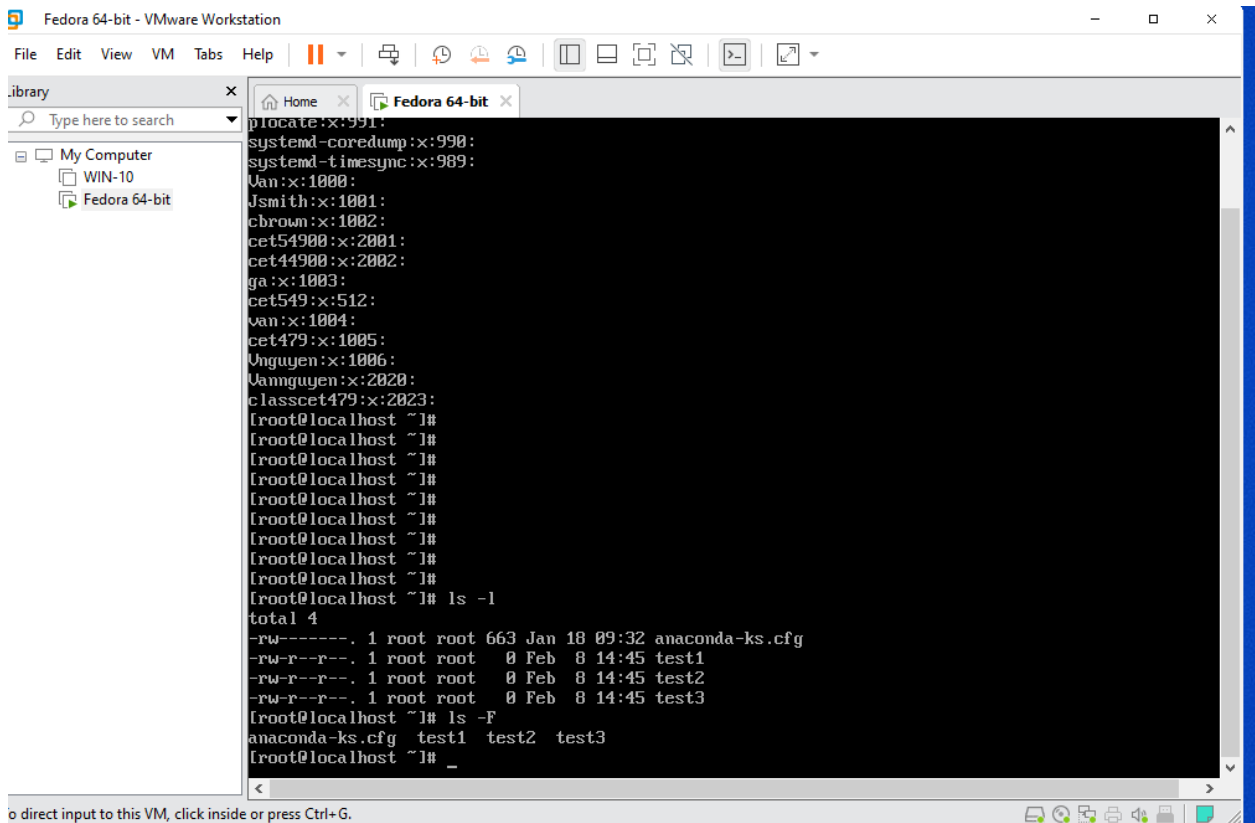
13. Display the use of ? wildcard in your Linux environment and explain what you observed.

The wildcard characters representation is listed for informational purposes only:

- **Asterisk (*)**: matches any string, including an empty string.
- **Question mark (?)**: matches a single character.

14. What is the difference between ls -l and ls -F commands? What command can you use to learn about their description?

```
# [ls -l] : Show as a list  
  
# [ls -F] : list the contents of your current working  
directory
```



The screenshot shows a VMware Workstation window titled "Fedora 64-bit - VMware Workstation". The interface includes a menu bar (File, Edit, View, VM, Tabs, Help), a toolbar, and a sidebar with a "library" section showing "My Computer" and "Fedora 64-bit". The main area is a terminal window with the following output:

```
piocate:x:991:
systemd-coredump:x:990:
systemd-timesync:x:989:
van:x:1000:
Jsmith:x:1001:
cbrown:x:1002:
cet54900:x:2001:
cet44900:x:2002:
ga:x:1003:
cet549:x:512:
van:x:1004:
cet479:x:1005:
Unguyen:x:1006:
Vannguyen:x:2020:
classcet479:x:2023:
[root@localhost ~]#
[root@localhost ~]#
[root@localhost ~]#
[root@localhost ~]#
[root@localhost ~]#
[root@localhost ~]#
[root@localhost ~]#
[root@localhost ~]#
[root@localhost ~]#
[root@localhost ~]#
[root@localhost ~]# ls -l
total 4
-rw-----. 1 root root 663 Jan 18 09:32 anaconda-ks.cfg
-rw-r--r--. 1 root root  8 Feb  8 14:45 test1
-rw-r--r--. 1 root root  8 Feb  8 14:45 test2
-rw-r--r--. 1 root root  8 Feb  8 14:45 test3
[root@localhost ~]# ls -F
anaconda-ks.cfg test1 test2 test3
[root@localhost ~]# _
```

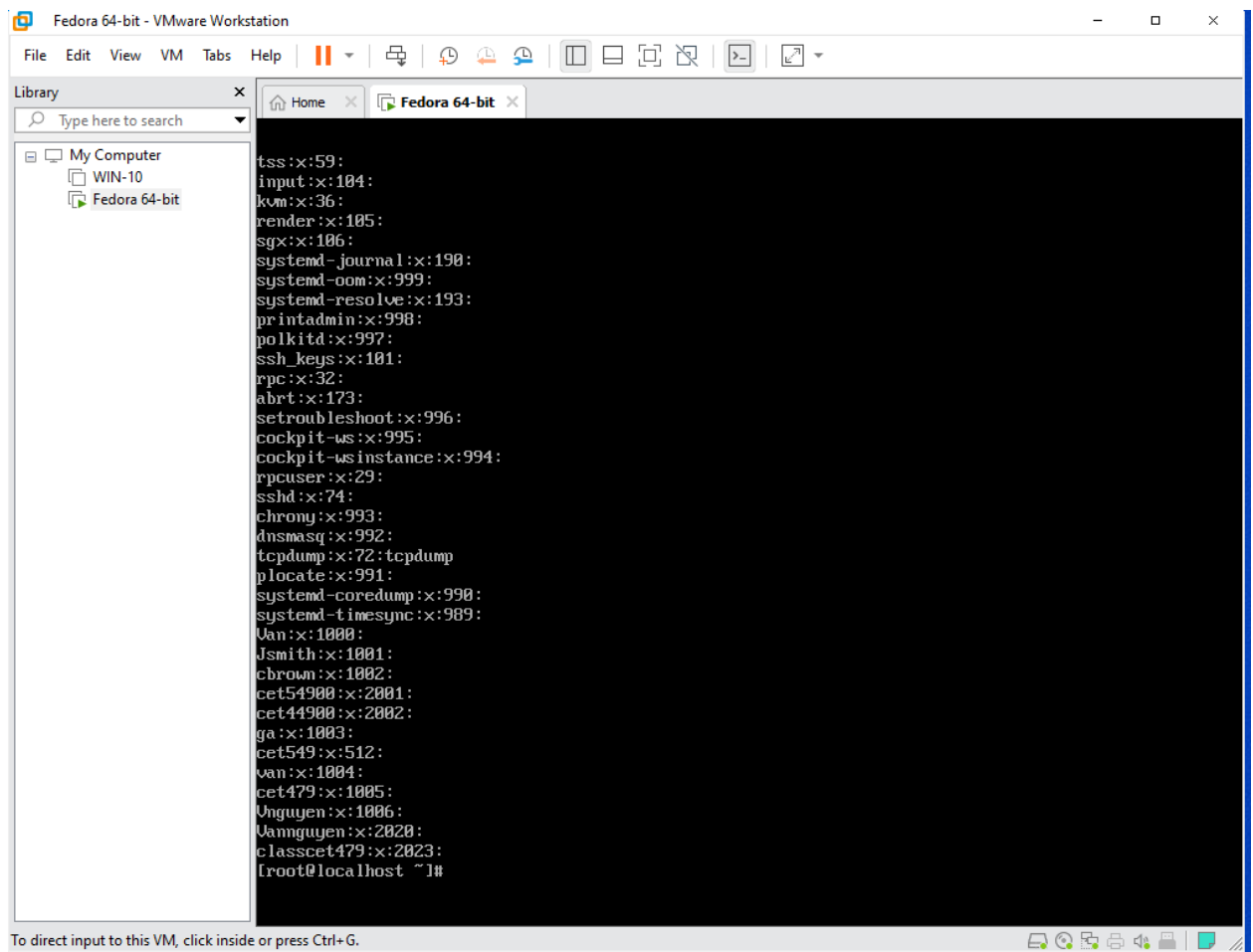
15. What kind of user properties can you change using the user mod command? Show an example in your Linux environment.

After creating a user, we must sometimes change their attributes like password or login directory etc. so to do that we use the **[usermod]** command. The information of a user is stored in the following files:

- /etc/passwd** : User account information.
- /etc/group** : Group account information.
- /etc/shadow** : Secure account information.
- /etc/login.defs** : Shadow password suite configuration.
- /etc/gshadow** : Secure group account information.

16. Display the list of all the groups in your home directory. What commands did you use?

[`cat /etc/group`] : Will be presented with the list of groups available on the system.



The screenshot shows a VMware Workstation window titled "Fedora 64-bit - VMware Workstation". The interface includes a menu bar (File, Edit, View, VM, Tabs, Help), a toolbar, and a Library pane on the left. The Library pane shows a tree view with "My Computer" expanded, containing "WIN-10" and "Fedora 64-bit". The main window displays a terminal window titled "Fedora 64-bit" with a black background and white text. The terminal output lists system and user groups with their IDs and passwords, such as "tss:x:59:", "input:x:104:", "kvm:x:36:", "render:x:105:", "sgx:x:106:", "systemd-journal:x:198:", "systemd-oom:x:999:", "systemd-resolve:x:193:", "printadmin:x:998:", "polkitd:x:997:", "ssh_keys:x:101:", "rpc:x:32:", "abrt:x:173:", "setroubleshoot:x:996:", "cockpit-ws:x:995:", "cockpit-ws instance:x:994:", "rpcuser:x:29:", "sshd:x:74:", "chrony:x:993:", "dnsmasq:x:992:", "tcpdump:x:72:tcpdump", "plocate:x:991:", "systemd-coredump:x:990:", "systemd-timesync:x:989:", "Uan:x:1000:", "Jsmith:x:1001:", "cbrown:x:1002:", "cet54900:x:2001:", "cet44900:x:2002:", "ga:x:1003:", "cet549:x:512:", "van:x:1004:", "cet479:x:1005:", "Unguyen:x:1006:", "Uannguyen:x:2020:", "classcet479:x:2023:", and "[root@localhost ~]#". At the bottom of the window, a status bar reads "To direct input to this VM, click inside or press Ctrl+G."

```
tss:x:59:
input:x:104:
kvm:x:36:
render:x:105:
sgx:x:106:
systemd-journal:x:198:
systemd-oom:x:999:
systemd-resolve:x:193:
printadmin:x:998:
polkitd:x:997:
ssh_keys:x:101:
rpc:x:32:
abrt:x:173:
setroubleshoot:x:996:
cockpit-ws:x:995:
cockpit-ws instance:x:994:
rpcuser:x:29:
sshd:x:74:
chrony:x:993:
dnsmasq:x:992:
tcpdump:x:72:tcpdump
plocate:x:991:
systemd-coredump:x:990:
systemd-timesync:x:989:
Uan:x:1000:
Jsmith:x:1001:
cbrown:x:1002:
cet54900:x:2001:
cet44900:x:2002:
ga:x:1003:
cet549:x:512:
van:x:1004:
cet479:x:1005:
Unguyen:x:1006:
Uannguyen:x:2020:
classcet479:x:2023:
[root@localhost ~]#
```

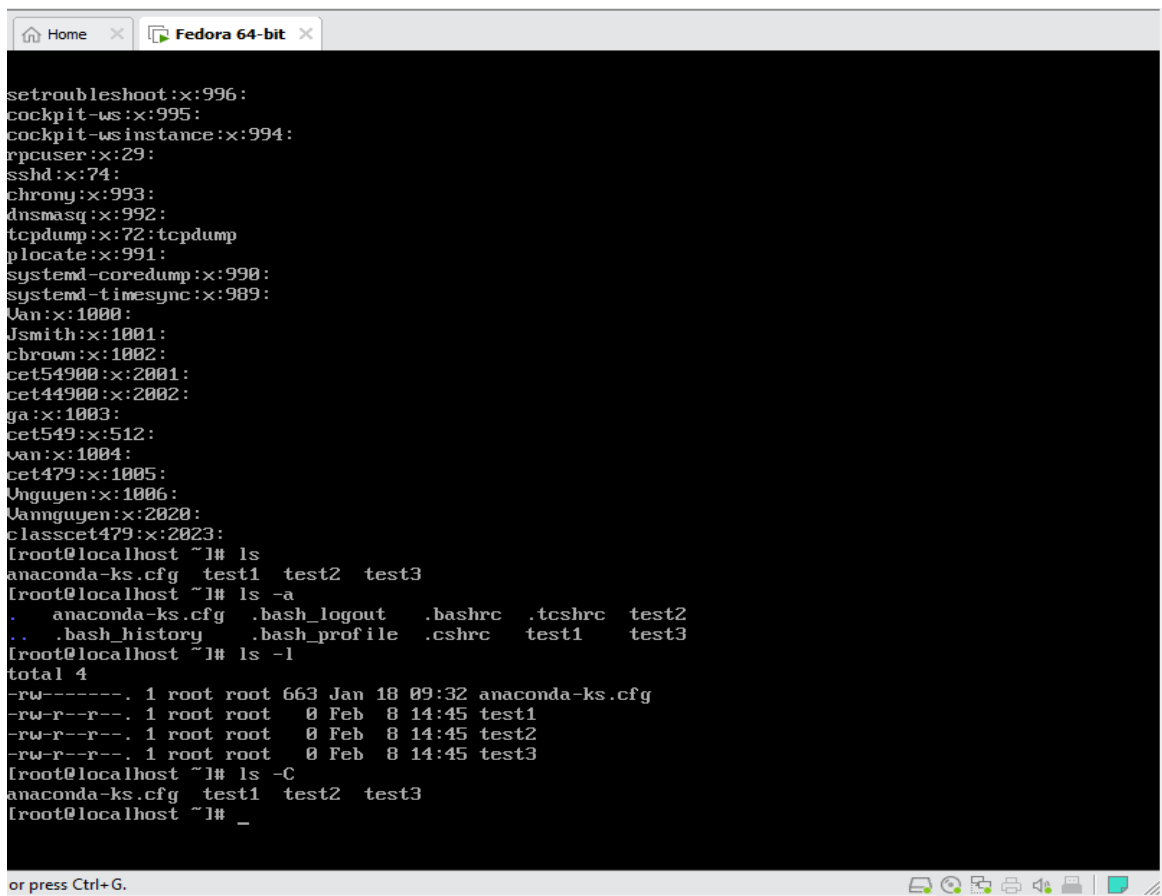
17. If you want to display the contents of your home directory, which command would you use? Check if your command (set of commands) works.

[**ls**] : Displays all information in alphabetic order by file name.

[**ls -a**] : To show all entries for files, including those that begin with a dot (.)

[**ls -l**] : Output list one entry per line.

[**ls -C or -x**] : List entries in multiple columns.



```
setroubleshoot:x:996:
cockpit-ws:x:995:
cockpit-wsinstance:x:994:
rpcuser:x:29:
sshd:x:74:
chrony:x:993:
dnsmasq:x:992:
tcpdump:x:72:tcpdump
plocate:x:991:
systemd-coredump:x:990:
systemd-timesync:x:989:
van:x:1000:
Jsmith:x:1001:
cbrown:x:1002:
cet54900:x:2001:
cet44900:x:2002:
ga:x:1003:
cet549:x:512:
van:x:1004:
cet479:x:1005:
Unguyen:x:1006:
Vannguyen:x:2020:
classcet479:x:2023:
[root@localhost ~]# ls
anaconda-ks.cfg  test1  test2  test3
[root@localhost ~]# ls -a
.  anaconda-ks.cfg  .bash_logout  .bashrc  .tcshrc  test2
.. .bash_history  .bash_profile .cshrc   test1   test3
[root@localhost ~]# ls -l
total 4
-rw-----. 1 root root 663 Jan 18 09:32 anaconda-ks.cfg
-rw-r--r--. 1 root root  0 Feb  8 14:45 test1
-rw-r--r--. 1 root root  0 Feb  8 14:45 test2
-rw-r--r--. 1 root root  0 Feb  8 14:45 test3
[root@localhost ~]# ls -C
anaconda-ks.cfg  test1  test2  test3
[root@localhost ~]# _
```

18. Submission Policy: Answers should be readable to score credits. Upload your answers and screenshots online using a word document.

